

A Proofs of Main Results

A.1 Proof of Proposition 1

Proof. Since $m \in C^{r+1}(\mathcal{Z})$ and $\mathcal{Z}$ is compact, every partial derivative of order at most $r + 1$ is continuous and therefore bounded. By the multivariate Stone–Weierstrass theorem, there exists a sequence of polynomials $\{P_r\}$ satisfying

$$\sup_{Z \in \mathcal{Z}} |m(Z) - P_r(Z)| \rightarrow 0.$$

Moreover, classical approximation theory for multivariate polynomials implies that whenever m possesses $(r + 1)$ continuous derivatives, the approximation error satisfies

$$\sup_{Z \in \mathcal{Z}} |m(Z) - P_r(Z)| \leq Ch_r^{r+1},$$

where the constant C depends only upon upper bounds of the derivatives of m . Defining $R_r(Z) = m(Z) - P_r(Z)$, immediately yields

$$\sup_{Z \in \mathcal{Z}} |R_r(Z)| \leq Ch_r^{r+1}.$$

Since $h_r \rightarrow 0$ as the approximation becomes increasingly rich, $\sup_{Z \in \mathcal{Z}} |R_r(Z)| \rightarrow 0$, which completes the proof. □

A.2 Proof of Proposition 2

Proof. Since $\mathcal{Z}$ is compact, the following holds as

$$|R_r(Z)|^q \leq \left(\sup_{u \in \mathcal{Z}} |R_r(u)| \right)^q.$$

Therefore, we can write furthermore that $\int_{\mathcal{Z}} |R_r(Z)|^q dZ \leq |\mathcal{Z}| \left(\sup_u |R_r(u)| \right)^q$. The uniform convergence established previously implies $\sup_u |R_r(u)| \rightarrow 0$, which gives $\|R_r\|_{L_q} \rightarrow 0$.

If Z possesses a density bounded on $\mathcal{Z}$, then $E\{R_r(Z)^2\} = \int_{\mathcal{Z}} R_r(z)^2 f_Z(z) dz$ is bounded above by $\|f_Z\|_{\infty} \int_{\mathcal{Z}} R_r(z)^2 dz$, which converges to 0. □

A.3 Proof of Theorem 1

Proof. Substituting the polynomial representation into the regression model gives

$$Y = m(Z) + \varepsilon = P_r(Z) + R_r(Z) + \varepsilon.$$

Since $\sup_Z |R_r(Z)| \rightarrow 0$, we have $R_r(Z) = o(1)$ uniformly over the support of Z . Hence, $Y = P_r(Z) + \varepsilon + o(1)$, which immediately implies $Y - P_r(Z) = \varepsilon + o(1)$. □

A.4 Proof of Proposition 3

Proof. Using the definition of the forward difference operator,

$$\Delta^d m(Z_i) = \sum_{j=0}^d (-1)^{d-j} \binom{d}{j} m(Z_{i+j}).$$

Since $m \in C^{d+1}(\mathcal{Z})$, a multivariate Taylor expansion of $m(Z_{i+j})$ around Z_i gives

$$m(Z_{i+j}) = \sum_{k=0}^d \frac{1}{k!} D^k m(Z_i) (Z_{i+j} - Z_i)^k + R_{ij},$$

where $|R_{ij}| \leq C \|Z_{i+j} - Z_i\|^{d+1}$. Substituting the expansion into the definition of $\Delta^d m(Z_i)$ yields

$$\Delta^d m(Z_i) = \sum_{k=0}^d \frac{D^k m(Z_i)}{k!} \sum_{j=0}^d (-1)^{d-j} \binom{d}{j} (Z_{i+j} - Z_i)^k + \sum_{j=0}^d (-1)^{d-j} \binom{d}{j} R_{ij}.$$

The combinatorial identity $\sum_{j=0}^d (-1)^{d-j} \binom{d}{j} j^k = 0$ for $k < d$ implies that every polynomial term of degree strictly smaller than d cancels exactly. Therefore, $\Delta^d m(Z_i) = O(h_n^d)$, uniformly over i . Since $h_n \rightarrow 0$, then $\sup_i |\Delta^d m(Z_i)| \rightarrow 0$, which completes the proof. $\square$

A.5 Proof of Corollary 1

Proof. Uniform convergence established in Proposition 3 immediately implies $\sup_i |\Delta^d m(Z_i)| \rightarrow 0$. Therefore, $\left| \Delta^d m(Z_i) \right|^2 \leq \left(\sup_i |\Delta^d m(Z_i)| \right)^2$, which converges to zero. Taking expectations preserves the limit. $\square$

A.6 Proof of Theorem 2

Proof. Since $Y_i = m(Z_i) + \varepsilon_i$, linearity of the difference operator gives

$$\Delta^d Y_i = \Delta^d m(Z_i) + \Delta^d \varepsilon_i.$$

By Proposition 3, $\Delta^d m(Z_i) = o(1)$ uniformly over i . Hence, $\Delta^d Y_i = \Delta^d \varepsilon_i + o(1)$.

Furthermore, $\Delta^d Y_i - \Delta^d \varepsilon_i = \Delta^d m(Z_i)$. Since the right-hand side converges uniformly to zero, it follows that $\left| \Delta^d Y_i - \Delta^d \varepsilon_i \right| \xrightarrow{P} 0$. Therefore, $\Delta^d Y_i = \Delta^d \varepsilon_i + o_P(1)$, establishing the desired stochastic approximation. $\square$

A.7 Proof of Proposition 4

Proof. Since $\Delta^d Y_i - \Delta^d \varepsilon_i = o_P(1)$, the Continuous Mapping Theorem yields

$$g(\Delta^d Y_i) - g(\Delta^d \varepsilon_i) = o_P(1).$$

Because g is bounded, the sequence is uniformly integrable. Therefore, by dominated convergence, $E[g(\Delta^d Y_i)] - E[g(\Delta^d \varepsilon_i)] \rightarrow 0$.

This establishes that every bounded continuous functional of the differenced responses converges to the corresponding functional of the latent error process.

□

A.8 Proof of Proposition 5

Proof. Since $m(Z) = P_r(Z) + R_r(Z)$, we have $P_r(Z) = m(Z) - R_r(Z)$. The polynomial approximation theorem established previously implies $R_r(Z) = o(1)$ uniformly over $\mathcal{Z}$. Likewise, $\Delta^d Y = \Delta^d \varepsilon + o_P(1)$. Hence,

$$\begin{aligned} (P_r(Z), \Delta^d Y) &= (m(Z) - R_r(Z), \Delta^d \varepsilon + o_P(1)) \\ &= (m(Z), \Delta^d \varepsilon) + (-R_r(Z), o_P(1)). \end{aligned}$$

Since $R_r(Z) = o(1)$, the joint remainder converges to zero in probability. Therefore,

$$(P_r(Z), \Delta^d Y) = (m(Z), \Delta^d \varepsilon) + o_P(1),$$

which proves the proposition.

□

A.9 Proof of Lemma 1

Proof. The forward difference operator is a measurable linear transformation of the error variables, i.e., $\Delta^d \varepsilon_i = \sum_{j=0}^d (-1)^{d-j} \binom{d}{j} \varepsilon_{i+j}$. Under the null hypothesis, $\{\varepsilon_i\}$ is independent of the covariate process $\{Z_i\}$. Independence is preserved under measurable transformations. Since $\Delta^d \varepsilon_i$ is a measurable function of the error variables alone, it follows immediately that $\Delta^d \varepsilon_i \perp\!\!\!\perp Z_i$.

□

A.10 Proof of Proposition 6

Proof. Since $P_r(Z) = m(Z) + o(1)$, the difference between the two random variables converges uniformly to zero. Consequently,

$$F_{P_r, \Delta^d \varepsilon}(u, v) - F_{m, \Delta^d \varepsilon}(u, v) \rightarrow 0$$

uniformly in (u, v) . Similarly, $F_{P_r}(u) - F_m(u) \rightarrow 0$. Therefore,

$$\begin{aligned} &F_{P_r, \Delta^d \varepsilon}(u, v) - F_{P_r}(u)F_{\Delta^d \varepsilon}(v) \\ &= F_{m, \Delta^d \varepsilon}(u, v) - F_m(u)F_{\Delta^d \varepsilon}(v) + o(1). \end{aligned}$$

Hence, $F_{P_r, \Delta^d \varepsilon} = F_{P_r}F_{\Delta^d \varepsilon}$ holds asymptotically if and only if $F_{m, \Delta^d \varepsilon} = F_mF_{\Delta^d \varepsilon}$.

□

A.11 Proof of Theorem 3

Proof. By Proposition 4, $P_r(Z) = m(Z) + o_P(1)$. Furthermore, $\Delta^d Y = \Delta^d \varepsilon + o_P(1)$.

Therefore, it is clear that $(P_r(Z), \Delta^d Y) = (m(Z), \Delta^d \varepsilon) + o_P(1)$. Since $\varepsilon \perp\!\!\!\perp Z$, Lemma 1 shows that $\Delta^d \varepsilon \perp\!\!\!\perp Z$. Because $m(Z)$ is a measurable function of Z , independence is preserved under measurable transformations, giving $m(Z) \perp\!\!\!\perp \Delta^d \varepsilon$.

By Proposition 5, $P_r(Z) \perp\!\!\!\perp \Delta^d \varepsilon$ asymptotically. Finally, we get $\Delta^d Y = \Delta^d \varepsilon + o_P(1)$, implies $P_r(Z) \perp\!\!\!\perp \Delta^d Y$ up to an asymptotically negligible error. Hence, $H_0 \iff H_0^{(r,d)}$ asymptotically. $\square$

A.12 Proof of Corollary 2

Proof. Theorem 3 establishes the asymptotic equivalence of the joint distributions of $(P_r(Z), \Delta^d Y)$ and $(m(Z), \Delta^d \varepsilon)$. Therefore, their empirical distribution functions differ by an $o_P(1)$ term uniformly over their support. Since the empirical Cramér–von Mises statistic is a continuous functional of the empirical process under the supremum norm, the continuous mapping theorem implies that replacing $(m(Z), \varepsilon)$ by $(P_r(Z), \Delta^d Y)$ changes the statistic only by an asymptotically negligible quantity. Hence, the proposed statistic has the same limiting behavior as the ideal statistic based on the latent errors. $\square$

A.13 Proof of Theorem 4

Proof. Under Assumption (A6), the indicator class is Donsker. Hence, $\sqrt{n}(F_n - F) \implies \mathbb{G}_F$, where $\mathbb{G}_F$ is the Brownian bridge indexed by rectangles. Under the null hypothesis,

$$F(u, w) = F_U(u)F_W(w).$$

Since multiplication is Hadamard differentiable, the functional delta method yields

$$\sqrt{n}(F_n - F_{n,U}F_{n,W}) \implies \mathbb{G}.$$

The covariance kernel follows directly from the covariance structure of the Brownian bridge under independence.

Finally, Assumptions (A2)–(A3) imply that replacing (m, ε) by $(P_r, \Delta^d Y)$ introduces an $o_P(1)$ perturbation, which does not affect weak convergence by Slutsky’s theorem. $\square$

A.14 Proof of Theorem 5

Proof. By the previous theorem, $\mathbb{G}_n \implies \mathbb{G}$. The mapping $x \mapsto \int x^2 dF$ is continuous on $\ell^\infty(\mathbb{R}^2)$. Hence, the Continuous Mapping Theorem gives

$$T_{n,r,d} \implies \int \mathbb{G}^2 dF.$$

Replacing F_n by F inside the integral changes the statistic only by $o_P(1)$, which follows from the Glivenko–Cantelli theorem. $\square$

A.15 Proof of Theorem 6

Proof. Under the alternative, $F \neq F_U F_W$. The uniform law of large numbers gives $F_n \rightarrow F$, uniformly. Therefore, $\frac{T_{n,r,d}}{n} = \int (F_n - F_{n,U} F_{n,W})^2 dF_n + o_P(1) \rightarrow \Delta$ as $n \rightarrow \infty$.

Since independence fails, we conclude that $\Delta > 0$. Hence, $T_{n,r,d} = n\Delta + o_P(n)$, which diverges to infinity, implying consistency. $\square$

A.16 Proof of Theorem 7

Proof. Let $\mathbb{G}_n(u, w) = \sqrt{n} \{F_n(u, w) - F_{n,U}(u)F_{n,W}(w)\}$ denote the empirical process corresponding to the transformed variables $U = P_r(Z)$ and $W = \Delta^d Y$. Under Assumptions (A1)–(A8), Theorem 4 establishes that $\mathbb{G}_n \rightsquigarrow \mathbb{G}$ in the space $\ell^\infty(\mathcal{U} \times \mathcal{W})$, where $\mathbb{G}$ is a centred Gaussian process with covariance function given in the theorem. The bootstrap sample $(U_1^*, W_1^*), \dots, (U_n^*, W_n^*)$ is generated independently from the empirical marginal distributions under the null hypothesis. Define the bootstrap empirical process

$$\mathbb{G}_n^*(u, w) = \sqrt{n} \{F_n^*(u, w) - F_{n,U}^*(u)F_{n,W}^*(w)\},$$

where F_n^* , $F_{n,U}^*$ and $F_{n,W}^*$ denote the bootstrap empirical distribution functions.

Since the bootstrap observations are sampled from the empirical distribution, the conditional empirical process satisfies the classical bootstrap functional central limit theorem. Consequently,

$$\mathbb{G}_n^* \rightsquigarrow \mathbb{G} \text{ conditionally in probability.}$$

Equivalently, $\sup_{h \in BL_1} |E^* h(\mathbb{G}_n^*) - E h(\mathbb{G})| \xrightarrow{P} 0$, where BL_1 denotes the class of bounded Lipschitz functions. The proposed statistic may be written as the functional $T_{n,r,d} = \Phi(\mathbb{G}_n)$, where $\Phi(f) = \int f^2(u, w) dF(u, w)$. Likewise, $T_{n,r,d}^* = \Phi(\mathbb{G}_n^*)$.

Since the mapping $f \mapsto \int f^2 dF$ is continuous on $\ell^\infty(\mathcal{U} \times \mathcal{W})$, the Continuous Mapping Theorem implies $T_{n,r,d} \rightsquigarrow \Phi(\mathbb{G})$ and, conditionally, $T_{n,r,d}^* \rightsquigarrow \Phi(\mathbb{G})$.

Applying the bootstrap version of the Continuous Mapping Theorem yields

$$\sup_t |P^*(T_{n,r,d}^* \leq t) - P(\Phi(\mathbb{G}) \leq t)| \xrightarrow{P} 0.$$

Since $T_{n,r,d} \rightsquigarrow \Phi(\mathbb{G})$, we also have $P(T_{n,r,d} \leq t) \rightarrow P(\Phi(\mathbb{G}) \leq t)$ at every continuity point t of the limiting distribution. Combining the preceding convergence results and using the triangle inequality gives $\sup_t |P^*(T_{n,r,d}^* \leq t) - P(T_{n,r,d} \leq t)| \xrightarrow{P} 0$.

Hence, the bootstrap consistently estimates the null distribution of the proposed Cramér–von Mises statistic, thereby providing asymptotically valid critical values and p -values for the proposed independence test. $\square$

A.17 Proof of Theorem 8

Proof. Under the contiguous alternatives, $\mathbb{G}_n \Rightarrow \mathbb{G} + \theta\Psi$, and therefore, by the Continuous Mapping Theorem, $T_{n,r,d} \Rightarrow T_\theta$. Consequently,

$$\lim_{n \rightarrow \infty} P_{H_n}(T_{n,r,d} > c_\alpha) = P(T_\theta > c_\alpha) = \beta(\theta).$$

When $\theta = 0$, the limiting distribution coincides with the null distribution. Hence,

$$\beta(0) = P(T_0 > c_\alpha) = \alpha.$$

For every $\theta > 0$, the limiting statistic contains the additional deterministic component

$$\theta^2 \int \Psi^2(u, w) dF(u, w),$$

which is strictly positive whenever independence fails. Accordingly, the distribution of T_θ is shifted to the right of the null distribution, implying $\beta(\theta) > \alpha$. Moreover, increasing θ increases the magnitude of the positive drift without affecting the covariance structure of the Gaussian component. Hence, the rejection probability is monotone increasing in θ . Finally, as $\theta \rightarrow \infty$, the deterministic drift dominates the stochastic fluctuations, yielding

$$T_\theta \xrightarrow{P} \infty.$$

Therefore, $\lim_{\theta \rightarrow \infty} P(T_\theta > c_\alpha) = 1$. This establishes all four assertions. $\square$

A.18 Proof of Theorem 9

Proof. Under the contiguous alternatives, both tests possess limiting distributions consisting of a centered Gaussian process together with a deterministic drift proportional to their respective efficacy parameters.

Since the local asymptotic power depends only upon the magnitude of the deterministic drift, the asymptotic sample sizes required by the two tests to attain identical local power satisfy $\frac{n_2}{n_1} \rightarrow \frac{\Delta_{\text{CVM}}}{\Delta_0}$. Therefore,

$$\boxed{\text{ARE} = \lim_{n \rightarrow \infty} \frac{n_2}{n_1} = \frac{\Delta_{\text{CVM}}}{\Delta_0}},$$

which immediately establishes the three cases stated above. $\square$